20XC87 – Mobile Security Lab

Project Aeterna: The Silas Vane Incident

22PC20 - N. Muthu Meenakshi

22PC38 – Vishnu Priya M

Flag 1: The Darkroom Protocol

"The system is blinded by the light of this world. It refuses to speak until it is returned to the void from which it came. Total absence is the only acceptable synchronization."

- **Clue:** The interface is waiting for a specific environment. Look into the device's sensory perception. What happens when the value of "external radiance" reaches absolute zero?

Flag 2: The Algebraic Cipher

"A name is just a sequence of weights. I've forgotten the word, but I remember the math. The seventh step is always seven times harder than the first."

- **Clue:** The gate is locked by a 11-digit sequence. Study the logic of the solveAlgebraicGate. The relationship between a character's position and its value is shifted by a constant factor of seven. Can you solve for X?

Flag 3: The Manifest Ghost

"They look at the code, but they forget the house that holds it. I've hidden a fragment in the legalities of the system, masked by a common standard of 64."

- **Clue:** Inspect the application's "blueprint" (the XML). There is an entry that defines a "Permission" that shouldn't exist. The name of this permission holds a string that looks like gibberish—unless you know how to decode a standard web-safe format.

Flag 4: The Deep Link Breach

"There is a back door for recovery, but there is no handle on the outside. You must call out to it by name, using a protocol from the year 2026."

- **Clue:** An Activity exists in the shadows. It is "Exported" to the world but has no UI trigger. You must use a system-level command to "View" a specific URI scheme hidden in the Manifest.

Flag 5: The Native Vault

"The heart of the engine is written in a language of metal (C++). It ignores the standard requests. It only yields to those who carry the 'Dead' signature found in the memory leak."

- **Clue:** The "Master Key" is behind a JNI bridge. The function expects a specific Hexadecimal integer. If you don't send the exact 4-byte key found in Fragment 6, the vault stays locked.

You'll need a "Toolkit" (Frida) to perform the injection.

Flag 6: The Forensic Trace

"Every soul leaves a digital footprint. Even hidden files leave a dot. I've shifted the memory so it's unreadable to the casual observer."

- **Clue:** There is a hidden file in the app's internal storage directory. Its name starts with a period. The contents are garbled—every character is exactly one step away from its true identity. Can you shift them back?

Flag 7: The Appended Asset

"I've hidden a secret in my portrait. Most people see the image; the wise look at the bytes at the very end of the line."

- **Clue:** Look into the assets folder. One of the visual files is "heavier" than it should be. The data isn't in the pixels; it's appended to the "tail" of the binary data. Use a tool that extracts readable strings from binary files.

Flag 8: The Developer's Secret Handshake

"Seven taps on the corrupted soul. No sound, no flash, just a silent entry into the logs. The final truth is written in the 'Vane Logs' of the shared system."

- **Clue:** There is a hidden counter on a specific piece of text in the Glitch UI. Tapping it repeatedly triggers a one-time write to the SharedPreferences. You won't see it happen, but the XML file will appear in the internal folders once the ritual is complete.